## 0.1 Hilbert Space

**Definition 0.1.1.** Complete Inner product Vector Space is called *Hilbert Space*.

### 0.1.1 Hilbert Space in $\mathbb{R}^\omega$

**Definition 0.1.2.** Define $\mathbb{R}^\omega \overset{\text{def}}{=} \prod_{i=1}^{\infty} \mathbb{R}$ as a countable product of Euclidean space $\mathbb{R}$ with product topology.
And define $\mathbb{H} \overset{\text{def}}{=} \left\{ \{x_n\}_{n=1}^{\infty} \;\middle|\; \sum_{n=1}^{\infty} x_n^2 < \infty \right\} \subset \mathbb{R}^\omega$, *Metric* on $\mathbb{H}$ as $\mu : \mathbb{H} \times \mathbb{H} \to \mathbb{R} : (\{x_n\}, \{y_n\}) \mapsto \sqrt{\sum_{i=1}^{\infty} (x_i - y_i)^2}$.
The Metric Space $(\mathbb{H}, \mu)$ is called *Hilbert Space* or $l_2$ *Space*.

Define the operations elementwise; then $(\mathbb{H}, +, \times)$ is a Vector Space over $\mathbb{R}$.
Moreover, $\mathbb{H}$ is Complete Metric Space and Inner product Vector Space.

**Lemma 0.1.3.** $\mu : \mathbb{H} \times \mathbb{H} \to \mathbb{R} : (\{x_n\}, \{y_n\}) \mapsto \sqrt{\sum_{i=1}^{\infty} (x_i - y_i)^2}$ is Metric function induced by the inner product.

*Proof.* We know that $\mathbb{R}^\omega$ is Vector Space. Moreover, $\mathbb{H} \subset \mathbb{R}^\omega$ is Subspace. Using subspace criteria:

$S \subset V$ is Subspace of Vector Space $V$ if and only if $0 \in S$ and For any $x, y \in S$ and $a \in F$, $ax + y \in S$.

Clearly, $\{0\} \in \mathbb{H}$. Let $a \in \mathbb{R}$ and $\{x_n\}, \{y_n\} \in \mathbb{H}$ be given. Then, $a\{x_n\} + \{y_n\} = \{ax_n + y_n\} \in \mathbb{H}$ because:

$$\sum_{i=1}^{\infty} (ax_i + y_i)^2 = \sum_{i=1}^{\infty} \left[ a^2 x_i^2 + 2a x_i y_i + y_i^2 \right] \overset{(*)}{=} a^2 \sum_{i=1}^{\infty} x_i^2 + 2a \sum_{i=1}^{\infty} x_i y_i + \sum_{i=1}^{\infty} y_i^2 < \infty$$

The $(*)$ given by:

$$\sum_{i=1}^{\infty} |x_i y_i| = \sum_{i=1}^{\infty} |x_i||y_i| \le \sum_{i=1}^{\infty} (\max(|x_i|, |y_i|))^2 \le \sum_{i=1}^{\infty} (x_n^2 + y_n^2) = \sum_{i=1}^{\infty} x_n^2 + \sum_{i=1}^{\infty} y_n^2 < \infty \tag{$*$}$$

Thus $\mathbb{H}$ is Vector Space over $\mathbb{R}$. Now, define *inner product* on $\mathbb{H}$ as:

$$\langle \;\cdot\;, \;\cdot\; \rangle : \mathbb{H} \times \mathbb{H} \to \mathbb{R} : (\{x_n\}, \{y_n\}) \mapsto \sum_{i=1}^{\infty} x_i y_i$$

This definition is well-defined since $(*)$. And, Linearity in first:

$$\langle a\{x_n\} + \{y_n\}, \{z_n\} \rangle = \langle \{ax_n + y_n\}, \{z_n\} \rangle = \sum_{i=1}^{\infty} (ax_i + y_i) z_i = a \sum_{i=1}^{\infty} x_i z_i + \sum_{i=1}^{\infty} y_i z_i = a\langle \{x_n\}, \{z_n\} \rangle + \langle \{y_n\}, \{z_n\} \rangle$$

The other conditions are clear. Thus, $(\mathbb{H}, \langle \cdot, \cdot \rangle)$ is *inner product space*.
Using *ineer product*, define the *Norm* on $\mathbb{H}$ as:

$$\| \cdot \| : \mathbb{H} \to \mathbb{R} : \{x_n\} \mapsto \sqrt{\langle \{x_n\}, \{x_n\} \rangle}$$

Finally, define *Metric* on $\mathbb{H}$ as:

$$\mu : \mathbb{H} \times \mathbb{H} \to \mathbb{R} : (\{x_n\}, \{y_n\}) \mapsto \|\{x_n\} - \{y_n\}\| = \sqrt{\sum_{i=1}^{\infty} (x_i - y_i)^2}$$

□

**Theorem 0.1.4.** *Hilbert Space* is Separable.

*Proof.* For each $n \in \mathbb{N}$, define $D_n \overset{\text{def}}{=} \{\{p_n\} \mid p_i \in \mathbb{Q},\ p_{n+1} = p_{n+1} = \cdots = 0\}$ and $D \overset{\text{def}}{=} \bigcup_{n\in\mathbb{N}} D_n$.

Then, $D$ is countable set. We will show that $\overline{D} = \mathbb{H}$.
Let $\varepsilon > 0$ and $\{x_n\} \in \mathbb{H}$ be given. Since convergence, there exists $N \in \mathbb{N}$ such that

$$\sum_{i=N+1}^{\infty} x_i^2 = \sum_{i=1}^{\infty} x_i^2 - \sum_{i=1}^{N} x_i^2 < \frac{\varepsilon^2}{2}$$

Since density of Rationals, put each $i = 1, 2, \ldots, N$, $p_i \in \mathbb{Q}$ $|x_i - p_i| < \dfrac{\varepsilon}{\sqrt{2N}}$ and $p_i = 0$ for $i \geq N+1$.
Then, $\{p_n\} \in D_n \subset D$ and

$$\mu(\{x_n\}, \{p_n\}) = \sqrt{\sum_{i=1}^{N}(x_i - p_i)^2 + \sum_{i=N+1}^{\infty}(x_i - p_i)^2} = \sqrt{\sum_{i=1}^{N}(x_i - p_i)^2 + \sum_{i=N+1}^{\infty} x_i^2} < \sqrt{N \cdot \frac{\varepsilon^2}{2N} + \frac{\varepsilon^2}{2}} = \varepsilon$$

□

**Corollary 0.1.5.** *Hilbert Space* is Second-Countable.

**Theorem 0.1.6.** *Hilbert Space* is Complete.

*Proof.* Let $\{\{x_{n,i}\}_{i=1}^{\infty}\}_{n=1}^{\infty}$ be a Cauchy sequence in $\mathbb{H}$. For any fixed $n, m \in \mathbb{N}$ and for each $j \in \mathbb{N}$,

$$|x_{n,j} - x_{m,j}| < \mu(\{x_{n,i}\}, \{x_{m,i}\}) = \sqrt{\sum_{i=1}^{\infty}(x_{n,i} - x_{m,i})^2}$$

That is, for each $j \in \mathbb{N}$, $\{x_{n,j}\}$ is Cauchy sequence in $\mathbb{R}$. Since $\mathbb{R}$ is Complete, put $y_j \overset{\text{def}}{=} \lim_{n\to\infty} x_{n,j}$, for each $j \in \mathbb{N}$.
Let $\varepsilon > 0$ be given. Then, there exists $N \in \mathbb{N}$ such that $n, m \geq N \implies \mu(\{x_{n,i}\}, \{x_{m,i}\}) < \dfrac{\varepsilon}{2}$.
Meanwhile, for each $k \in \mathbb{N}$,

$$\sum_{i=1}^{k}(x_{n,i} - x_{m,i})^2 \leq \sum_{i=1}^{\infty}(x_{n,i} - x_{m,i})^2 = [\mu(\{x_{n,i}\}, \{x_{m,i}\})]^2$$

Thus, $n, m \geq N \implies \displaystyle\sum_{i=1}^{k}(x_{n,i} - x_{m,i})^2 < \left(\frac{\varepsilon}{2}\right)^2$, for each $k \in \mathbb{N}$.

Taking limit to $m$, then $n \geq N \implies \displaystyle\lim_{m\to\infty}\left(\sum_{i=1}^{k}(x_{n,i} - x_{m,i})^2\right) = \sum_{i=1}^{k}\left(x_{n,i} - \lim_{m\to\infty} x_{m,i}\right)^2 = \sum_{i=1}^{k}(x_{n,i} - y_i)^2 < \left(\frac{\varepsilon}{2}\right)^2$.

And, for all $k \in \mathbb{N}$,

$$\sum_{i=1}^{k} y_i^2 = \sum_{i=1}^{k}(2(x_{n,i}^2 + (x_{n,i} - y_i)^2)) \leq 2\|\{x_{n,i}\}_{i=1}^{\infty}\|^2 + \left(\frac{\varepsilon}{2}\right)^2$$

Thus $\{y_i\} \in \mathbb{H}$. As a result,

$$n \geq N \implies \mu(\{x_n\}, \{y_n\}) = \sqrt{\sum_{i=1}^{\infty}(x_{n,i} - y_i)^2} = \sqrt{\lim_{k\to\infty}\sum_{i=1}^{k}(x_{n,i} - y_i)^2} < \frac{\varepsilon}{2}$$

□